

Heat Transfer to a Fluid in a Stirred Tank

Final Lab Report 2

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Unit Operations CHE 382

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Group 1, Thursday, AM section

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1. Executive Summary

Each of the experiments, both A and B, had a change in the impeller speed and a change in the cooling water rate. Also, for both experiments, while one aspect was being changed the other remained constant. The results section shows there was a constant increase in the heat transfer coefficient with or without the baffles. The first time around for experiment A three different impeller speeds were tested; 600 RPM, 800 RPM and 1000 RPM. Going from 600 RPM to 800 RPM there was an increase of 19.75% in the heat transfer coefficient and going from 800 RPM to 1000 RPM there was an increase of 16.1% in the heat transfer coefficient. The second time around for experiment A three different cooling water rates were tested, while keeping the impeller speed at 600 RPM. Between the first trial with temperature at 43.33 °C and the second trial with temperature at 48.89 °C there was an increase of 1.48% in the heat transfer coefficient. Between the second and third trial, 48.89 °C and 51.11 °C respectively, there was an increase of 1.76% increase in the heat transfer coefficient.

In comparison, for experiment B containing the baffles, the first time around going from 600 RPM to 800 RPM the increase of the heat transfer coefficient was at 21.12% and going from 800 RPM to 1000 RPM had an increase for the heat transfer coefficient of 16.1%. The second time around with the changing cooling water rate and a constant impeller speed of 600 RPM it had an increase of the heat transfer coefficient being 1.77% going from 43.33 °C to 48.89 °C and an increase of heat transfer coefficient being 0.58% going from 48.89 °C to 51.11 °C.

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As seen by the results this experiment, the experimental values are fairly close and agree with the theory. The theory being as the impeller speed increases so does the heat transfer coefficient.

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2. Introduction

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The objective of this experiment is to measure heat transfer coefficient between water and the inside of a tank wall and how the insertion of baffles affects it. Furthermore, effects of measured heat transfer coefficient on the impeller speed, the presence of baffles, and the fluid properties that are highly dependent on temperature are examined for this experiment. Heat transfer coefficient is defined for the difference in temperature between the inside vessel wall and the bulk mean temperature of the well stirred fluid. This experiment is significant because there are no experimental values for heat transfer coefficients in baffled, stirred tanks. With experimental values, processes can be designed more specifically with baffled tank heat transfer coefficients. An analysis of heat transfer coefficient will enable chemical engineers to control temperature throughout the process and hence it is essential to any industrial plant that is interested in precision temperature control.

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This experiment is done in two sections where heat transfer in an agitated tank was examined with baffle and without the baffle. The two parts of the experiment were conducted under the same sets of operating conditions varying only by the presence of baffles. The stirred tank in this experiment is composed of components such as an internal cooling coil, heat exchanger, steam condenser, pump, multiple thermocouples, and flowmeters. The heat exchanger was connected to the pump and water was continuously taken from the tank and cooled down through contact with a cool water stream. The water was then recycled back into the tank at a lower temperature. The flow rate of such recycle stream was manually manipulated by the flowmeter and an attached valve. The steam condenser continually condensed steam that

was used to heat the steam jacket. A stream of steam was flowing into the steam jacket at a constant rate throughout the entire experiment. The cooling coils inside the tank had a constant flow of cool water throughout them and such flow rate was always kept constant. Baffles are essentially two long, thin, and rectangular pieces of metal that are joined at a right angle and are placed on the inner edges of the stirred tank. Baffles enhance mixing and in theory increase heat transfer. Once steady state was achieved, indicated by the unfluctuating temperature change of the heat exchange inlet and outlet, temperatures of the cooling water, internal vessel, heat exchanger, and steam trap were recorded.

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An agitated tank in industry is used for a variety of different processes such as chemical reactions, blending, dispersions and leaching. A usual system has a stirred tank and either a steam jacket to give heat or an internal cooling coil. It can be necessary to add heat to a system to avoid phase changes or cool the system in case of an exothermic polymerization reaction. Furthermore, stirred tanks can be used to continually mix a catalyst and its reactant to improve a chemical reaction or used to extract a substance from a solid by mixing it with a liquid. To further the mixing process, baffles are often used, and an additional vessel fluid cooling can be achieved by a recirculation. Therefore, heat transfer in the agitated vessel plays an important role in maintaining the desired operating temperature of the mixing process or chemical reaction without changing phase.

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3. Theory

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Heat transfer coefficients must be experimentally determined given the complexity of the flow pattern within the agitated vessel. From experimental variables and apparatus geometry, dimensionless numbers can be calculated that will lead to an accurate correlation with the heat transfer coefficient (lab manual). This experiment uses a heat exchanger to cool down the water extracted from the stirred tank and returns it to cool down the system. As a result, it is reasonable to assume that convection is the dominant contributor to the heat transfer. Hence, transport dimensionless parameters such as Reynolds number (Re), Prandtl number (Pr), and Nusselt number can be calculated to determine the overall heat transfer coefficient between the fluid and the vessel wall.

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Reynolds Number:

The Reynolds number is defined in terms of impeller diameter and impeller speed. The fluid properties used to evaluate the subsequent groups corresponds to the fluid through which heat transport is taking place. From the equation, Reynolds number is proportional to impeller speed.

$$Re = \frac{D^2 n \rho}{\mu} \quad (1)$$

D [=] m (Diameter of the impeller)

n [=] rev/s (Revolutions per unit time of the impeller)

ρ [=] kg/m³ (Density of the fluid)

μ [=] kg/m*s (Viscosity of the fluid)

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Prandtl Number:

Heat capacity of the water in the tank is assumed to remain constant. As the temperature of the water increases, viscosity and thermal conductivity will both decrease.

$$Pr = \frac{Cp\mu}{k} \quad (2)$$

Cp [=] kJ/kg*K (Heat capacity of water)

k [=] W/m*K (Thermal conductivity of the fluid)

μ [=] kg/m*s (Viscosity of the fluid)

The Nusselt number is a function of the Reynolds and Prandtl numbers.

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(Inner wall, unbaffled, turbulent flow and Newtonian fluid.)

$$Nu = 0.36 \left(R_e^{\frac{2}{3}} \right) * \left(Pr^{\frac{1}{3}} \right) \quad (3)$$

Reynolds number will also increase as impeller speeds increase. Such increase holds a larger weight of the overall Nusselt number and the Prandtl number increases with an increased in temperature. Thus, the heat transfer coefficient at the inner surface of the agitated vessel can be calculated using the following equation:

$$h_{i,wall} = \frac{Nu * k}{D_T} \quad (4)$$

Finally, the overall heat transfer coefficient between the fluid and the vessel wall is calculated using the following equation:

$$U_i = \frac{1}{\left(\frac{1}{h_i} \right) + \left(\frac{1}{h_c} \right)} \quad (5)$$

h_c [=] W/m²*K (Heat transfer coefficient at the cooling coil)

h_i [=] W/m²*K (Heat transfer coefficient at the inner vessel wall)

In this experiment, the wall thickness is much smaller than the diameter of the tank.

Therefore, the overall heat transfer coefficient can be used to calculate the energy transferred between the vessel and the tank.

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$$Q_{13} = U_{i,wall}(t_1 - t_3)A \quad (6)$$

t_1 [=] K (Steam temperature in the jacket)

t_3 [=] K (Bulk average temperature of the fluid in the tank)

A [=] m²(Area of heat transfer surface)

In principle, when baffles are added to the tank, they would create more turbulent flow within the tank that allows the water to mix more effectively thereby speeding up the heat transfer process. As a result, the heat transfer coefficient is expected to increase when the baffles are added. In addition, since convection is affected by the medium's macroscopic speed, the heat transfer coefficient should also increase when the impeller speed is increased. However, there are no experimental values for heat transfer coefficients in a baffled agitated tank. Hence, the same equation can be used to calculate the heat transfer coefficient for the baffled tank.

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4. Results

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The goal of this experiment was to analyze the consequence of different agitation rates and its effect on the heat transfer coefficient. There were two experiments completed: experiment A and B. These experiments were analyzed to obtain the heat transfer between the fluid inside the vessel and the vessel was under certain conditions. Both experiment A and B were assumed to be at steady state with a continuous flow of water that was passing and leaving from the tank. With the addition and removal of baffles, the heat transfer of each experiment was also concluded.

In experiment A, three trials were performed at a cooling water rate of different percentages of the max flow rate of the right-hand meter of 1.1 GPM and impeller speeds of 600, 800, and 1000 RPM's without the baffles inside the vessel. Tables 1 and 3 were kept at constant flow rates, but the impeller speeds RPM's were changed from 600 to 1000. In table 1 (Unbaffled Vessel) from 610 to 800 RPM's, there is a 19.75 % increase of the heat transfer coefficient at the inner tank wall. Likewise, from 800 RPM to 1000 RPM, there is a 16.1% increase of the heat transfer coefficient. The difference between the two tables of 1 and 3 are the baffles inside the vessel. In experiment B and shown in table 3, with the baffles inside the vessel, from 600 RPM to 800 RPM, there is a 21.12% increase between the two RPM's. This means that the heat transfer coefficient between the two selected RPM's. Likewise, from 800 RPM to 1000 RPM with the baffles included, there is a 16.1% increase of the heat transfer coefficient between the two selected RPM's.

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In experiment A with the unbaffled, instead of changing the impeller speeds, the flowrate was changed while keeping the impeller speed constant at 600 RPM. In table 2, between the two temperatures of 43.33 °C at trial 1 and 48.89 °C at trial 2, there is a 1.48% increase of the heat transfer coefficient. Likewise, from trial 2 at 48.89 °C and trial 3 at 51.11 °C, there is a 1.76% increase of the heat transfer coefficient. Very similarly to experiment A, experiment B explores the use of baffles, but instead of changing the impeller speeds, once again, the flowrate was changed while keeping the impeller speeds constant. In table 4, trial 1 at 43.33 °C and trial 2 at 48.89 °C there is a 1.77% increase of the heat transfer coefficient. Likewise, from trial 2 at 48.89°C and trial 3 at 51.11 °C, there is a 0.58% increase of the heat transfer coefficient.

These experiments were concluded while keeping the level inside the vessel not exceeding 2 inches from the top. This was kept constant due to the potential overflow with the tank agitator. As shown in the previous experiments, parameters such as temperature, impeller speed, and the presence of baffles were documented as it took effect on the heat transfer coefficient.

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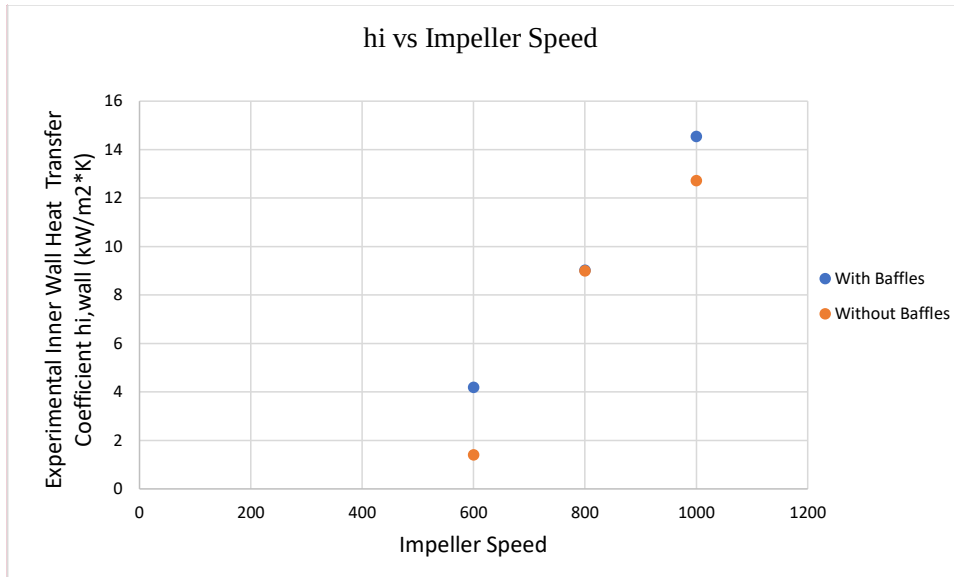


Figure 1: Effect of Baffles and Without Baffles on the Heat Transfer Coefficient

As shown in Figure 1, there is a correlation between the impeller speeds and the heat transfer coefficient. As impeller speed increase, the heat transfer increases for both with and without the baffles included inside the tank. Thus, making this a proportional relationship. Also shown in Figure 1, that with baffles inside the tank, it increases the heat transfer coefficient ultimately proving the theory behind this experiment. Baffles indeed do increase the heat transfer coefficient, due to the turbulent mixing inside the tank along with the tank agitator.

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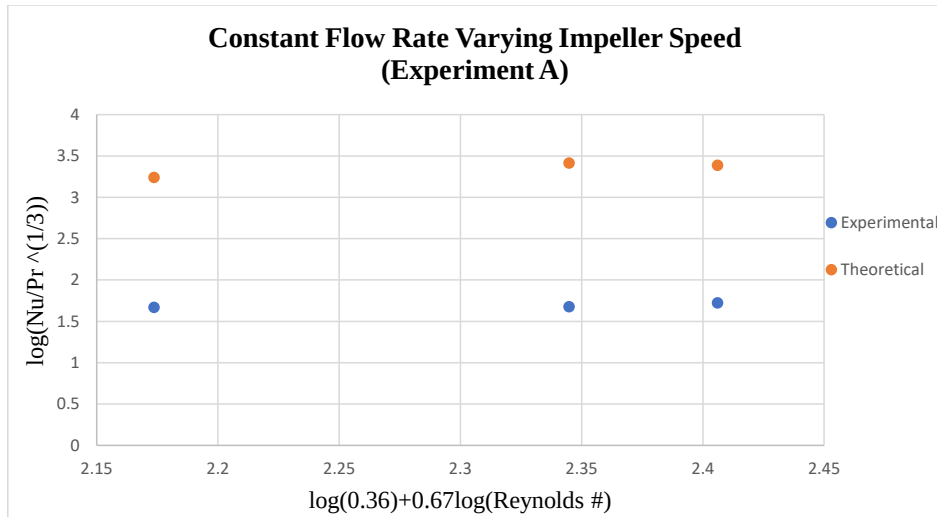


Figure 2: $\log(Nu/Pr^{1/3})$ vs. $\log(0.36)+0.67\log(\text{Reynolds \#})$ at Constant Flow Rate w/ Different Impeller Speeds ***EXPERIMENT A***

The correlations of heat transfer coefficients can be characterized by using the Prandtl number (Pr), Reynolds number (Re), and Nusselt number (Nu). Figure 2 depicts the correlative data between $\log(Nu/Pr^{1/3})$ and the $\log(0.36)+0.67\log(\text{Reynolds \#})$. Due to this correlation, the heat transfer coefficient for a unbaffled vessel follows the trend of the dimensionless groups. The trend is almost identical for experiment B, as it is shown in Figure 3.

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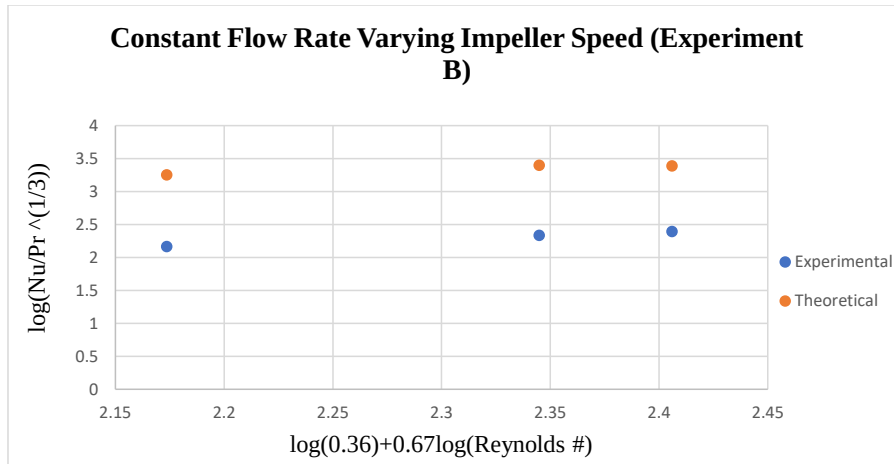


Figure 3: $\log(Nu/Pr^{1/3})$ vs. $\log(0.36)+0.67\log(\text{Reynolds \#})$ at Constant Flow Rate w/ Different Impeller Speeds ***EXPERIMENT B***

Table 1: Unbaffled Tank Heat Transfer Coefficients at Constant Temperature and Varying Impeller Speeds

Trial No.	Heat Transfer Coefficient at Inner Tank Wall ($\frac{W}{m^2 \cdot K}$)
1 (Tank = 37.78 °C, 610 rpm)	4961.130755
2 (Tank = 37.89 °C, 800 rpm)	5941.828845
3 (Tank = 37.78 °C, 1000 rpm)	6896.950431

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Table 2: Unbaffled Tank Heat Transfer Coefficients at Constant Impeller Speeds and Varying Temperature

Trial No.	Heat Transfer Coefficient at Inner Tank Wall $\left(\frac{W}{m^2 \cdot K}\right)$
1 (Tank = 43.33 °C, 600 RPM)	4743.64
2 (Tank = 48.89 °C, 600 RPM)	4813.687
3 (Tank = 51.11 °C, 600 RPM)	4898.634

Table 3: Baffled Tank Heat Transfer Coefficients at Constant Temperature and Varying Impeller Speeds

Trial No.	Heat Transfer Coefficient at Inner Tank Wall $\left(\frac{W}{m^2 \cdot K}\right)$
1 (Tank = 50 °C, 600 rpm)	4905.689292
2 (Tank = 49.9 °C, 800 rpm)	5941.828845
3 (Tank = 50.2 °C, 1000 rpm)	6896.950431

Table 4: Baffled Tank Heat Transfer Coefficients at Constant Impeller Speeds and Varying Temperature

Trial No.	Heat Transfer Coefficient at Inner Tank Wall $\left(\frac{W}{m^2 \cdot K}\right)$
1 (Tank = 43.33 °C, 600 RPM)	4797.36872
2 (Tank = 48.89 °C, 600 RPM)	4882.412875
3 (Tank = 51.67 °C, 600 RPM)	4910.856378

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5. Discussion

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In this experiment, the dependence of heat transfer coefficient on impeller speeds, fluid characteristics, and the presence of baffles were studied. The results that were obtained experimentally was evaluated as the system reached steady state. The heat exchanger flow rate, steam condensate flow rate, and the impeller speeds were all varied. Theoretically, at higher impeller speeds, the heat transfer coefficient should increase due to the efficient mixing. This enables for a larger area of the fluid to be exposed to the hot surface of the jacketed vessel. Similarly, to this theory, the baffled tank experiment (experiment B), it is anticipated that the heat transfer coefficient will be much higher than the unbaffled (experiment A) due to the mixing. This is because the baffles will cause turbulent flow within in itself while the tank agitator is mixing the fluid inside. Because of this, the heat transfer coefficient is greater than the unbaffled tank. This is shown in *Table 2* and *Table 4*, were the trials of 1 and 2 in *Table 2* show a 1.48% increase of the heat transfer coefficient compared in *Table 4* of trials 1 and 2, a 1.76% increase of the heat transfer coefficient. Shown in *Figure 1*, the relationship between an unbaffled and baffled tank, once again exhibits a higher heat transfer coefficient as the impeller speed increases. The theory is proved that with a baffled tank, heat transfer coefficients increase.

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Shown in *Figures 2 -3*, the relationship between the Reynolds number (Re) and the Nusselt (Nu)/Prandtl (Pr) numbers. Reynolds number in experiment B (baffled tank), shows an increasing behavior, while the Nusselt and Prandtl numbers follow with. This shows once again the theory behind this experiment where baffles inside the tank allow for greater mixing, thus, greater turbulent flow inside in the tank. Contrary to experiment B shown in *Figure 3*, *Figure 2*

shows a very constant, minimalistic change as impeller speeds increase. Reynolds number as a function of logarithmic stays steady throughout. This again proves the theory mention earlier.

There were several factors involved in this experiment that had a possibility of affected the results gathered. One factor is the difficulty achieving steady state in the system. When adjusting the flows of the system, it takes a long time for the instruments to read the correct measurements and temperature readings. Another challenge that interfered with the data collected is the digital reader of the impeller speed kept fluctuating. As shown in *Table 1*, trial 1 is at 610 RPM rather than 600 rpm. It was very difficult to constantly have 600 RPM as the reading fluctuated ± 25 RPM.

Commented [DL33]: Why there are differences between the theoretical and the experimental results? Prove a discussion on that, not only the error analysis.

Commented [DL34]: Good.

6. References

[1].“Properties of Water.” *Wikipedia*, Wikimedia Foundation, 3 Feb. 2020, en.wikipedia.org/wiki/Properties_of_water.

[2]Bird, Stewart, and Lightfoot. “Transport Phenomena.” John Wiley and Sons, NJ: 2007.

[3]Kay J M & Nedderman R M. “Fluid Mechanics and Transfer Processes.” Cambridge University Press: 1985

[4]. Thermal Kinetics. “Common Uses and Applications for Evaporators.” *Thermal Kinetics*, Thermal Kinetics, 23 Aug. 2019, thermalkinetics.net/common-uses-applications-evaporators.

[5] Lab Manual

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7. Appendix I: Data Tabulation/Graphs

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Table 5: Flowrates and Impeller Speeds

Flowrates and Impeller Speeds					
Trial No.	Recycled Flowrate %	Recycled Flowrate kg/s	Cooling Water Flowrate %	Cooling Water Flowrate kg/s	RPM
1	73	0.050661272	50	0.029652301	600
2	78	0.054131222	50	0.029652301	800
3	80	0.055519202	50	0.029652301	1000

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Table 6: Temperature of Experiment A w/ Baffles (Kelvin)

Changing Impeller Speeds with Baffles (Kelvin)								
Trial No.	RPM	T1 (Cooling Water In)	T2 (Cooling Water Out)	T3 (Hex In)	T4 (Hex Out)	T5 (Trap Cooling)	T6 (Steam Trap)	T7 (Tank)
1	600	276.483	293.15	323.15	317.65	279.817	277.594	323.15
2	800	276.483	292.594	319.15	313.65	278.706	277.594	318.706
3	1000	275.372	292.039	316.15	311.15	278.706	277.594	275.372

Table 7: Temperature of Experiment B w/o Baffles (Kelvin)

Changing Impeller Speeds Without Baffles (Kelvin)								
Trial No.	RPM	T1 (Cooling Water In)	T2 (Cooling Water Out)	T3 (Hex In)	T4 (Hex Out)	T5 (Trap Cooling)	T6 (Steam Trap)	T7 (Tank)
1	610	278.15	310.928	313.15	304.15	279.817	278.15	310.928
2	800	277.594	292.039	314.15	310.15	319.15	315.15	312.039
3	1000	276.483	292.039	314.15	309.15	279.817	277.594	310.928

Table 8: Experiment A w/ Baffles

Experiment A with Baffles							
Trial No.	Q (ΔH_{Ex} kJ/s)	Q Coils (kJ/s)	Q Overall (kJ/s)	U exp ($\frac{kW}{m^2 \cdot K}$)	Reynolds #	Prandtls #	Nusselts #
1	-1.166374465	2.068783575	0.902409109	1.218417499	14523.68775	3.435677	323.3637
2	-1.246263124	1.999770335	0.753507211	1.272935874	17921.31861	3.695705	381.1667
3	-1.162016898	2.068783575	0.906766677	2.276845704	8061.271093	10.31655	315.0767

Table 9: Experiment B w/o Baffles

Experiment A Without Baffles							
Trial No.	Q (ΔH_{Ex}) (kJ/s)	Q Coils (kJ/s)	Q Overall (kJ/s)	U exp $\left(\frac{kW}{m^2 \cdot K}\right)$	Reynolds #	Prandtls #	Nusselts #
1	-1.908612761	4.068553909	2.159941148	2.928666255	14523.68775	3.435677	323.3637
2	-0.906373181	1.792978865	0.886605683	1.269308754	17921.31861	3.695705	381.1667
3	-1.162016898	1.93088122	0.768864322	1.389905799	8061.271093	10.31655	315.0767

Table 10: Heat Transfer Coefficients for Experiment A w/ Baffles $\left(\frac{kW}{m^2 \cdot K}\right)$

Heat Transfer Coefficients Experiment A w/ Baffles						
Trial No.	hi experimental $\left(\frac{kW}{m^2 \cdot K}\right)$	h0 Experimental $\left(\frac{kW}{m^2 \cdot K}\right)$	hi Theoretical $\left(\frac{kW}{m^2 \cdot K}\right)$	h0 Theoretical $\left(\frac{kW}{m^2 \cdot K}\right)$	hi %error	h0 %error
1	0.943497634	4.181487682	10.91634087	11.73707761	1057.007763	180.6914
2	1.115487729	9.018488914	10.95149208	11.73707761	881.7671494	30.14461
3	0.915183508	1.530285297	11.29787349	11.73707761	1134.492689	666.9862

Table 11: Heat Transfer Coefficients for Experiment A w/o Baffles $\left(\frac{kW}{m^2 \cdot K}\right)$

Heat Transfer Coefficients Experiment A Without Baffles						
Trial No.	hi experimental $\left(\frac{kW}{m^2 \cdot K}\right)$	h0 Experimental $\left(\frac{kW}{m^2 \cdot K}\right)$	hi Theoretical $\left(\frac{kW}{m^2 \cdot K}\right)$	h0 Theoretical $\left(\frac{kW}{m^2 \cdot K}\right)$	hi %error	h0 %error
1	0.943497634	1.391916865	11.39562525	11.73707761	1107.806447	743.2312
2	1.112152938	8.982584872	10.94924723	11.73707761	884.5091314	30.66481
3	0.919318373	2.715257288	11.01760439	11.73707761	1098.453627	332.2639

Table 12: Experiment A w/ Baffles Dimensionless Variable Analysis

Experiment A w/ Baffles Dimensional Variable Analysis					
Trial No.	$\log(0.36) + 0.67 \log(\text{Reynolds \#})$	Nusselt Experimental	$\log(\text{experimental Nu/Pr}^{1/3})$	Nusselt # Theoretical	$\log(\text{theoretical Nu/Pr}^{1/3})$
1	2.344894026	323.363657	2.331020436	3741.342614	3.394356709
2	2.406060375	381.166659	2.391882475	3753.38994	3.385191255
3	2.173592863	315.0767317	2.160571518	3872.104769	3.250102286

Table 13: Experiment A w/o Baffles Dimensionless Variable Analysis

Experiment A Without Baffles Dimensional Variable Analysis					
Trial No.	Log (0.36) +0.67log(Reynolds #)	Nusselt Experimental	log (experimental Nu/Pr ^{1/3})	Nusselt # Theoretical	log (theoretical Nu/Pr ^{1/3})
1	2.344894026	323.363657	1.673127472	3905.607095	3.41301778
2	2.406060375	380.0271474	1.719442987	3752.620564	3.385102223
3	2.173592863	316.5002712	1.667568642	3776.048521	3.239192753

Table 14: Theoretical Values using Water Property Plots for Experiment A

Theoretical Values Using Water Property Plots for Experiment A w/ Baffles					
Trial No.	Tank (Bulk Temperature)	Prandtl #	Viscosity (kg/m*s)	Thermal Conductivity	Temperature (°C)
1	323.15	3.435676896	0.00054416	0.667	50
2	318.706	3.695704957	0.000587993	0.669	45.556
3	275.372	10.31654889	0.001633988	0.664	2.222

Table 15: Theoretical Values using Water Property Plots for Experiment B

Theoretical Values Using Water Property Plots for Experiment B Without Baffles					
Trial No.	Tank (Bulk Temperature)	Prandtl #	Viscosity (kg/m*s)	Thermal Conductivity	Temperature (°C)
1	310.928	3.435676896	0.00067994	0.667	37.778
2	312.039	3.695704957	0.000665439	0.669	38.889
3	310.928	10.31654889	0.00067994	0.664	37.778

8. Appendix II: Error Analysis

Experimental errors may have been caused by the equipment used in the experiment. While performing the experiment, the impeller speed was measured by reading a digital display. Once the appropriate impeller speed was reached, the value would not remain constant and would fluctuate ± 25 RPM. The uncertainty of the impeller speed may be found using the equation:

$$\% \text{ uncertainty} = \frac{\text{uncertainty}}{\text{RPM}} * 100 \quad (7)$$

Table 16: Uncertainty % For Table 1

Trial No.	RPM	Uncertainty	% Uncertainty
1	610	25	4.098360656
2	800	25	3.125
3	1000	25	2.5

By analyzing Table 16, as impeller speeds increase, the “% Uncertainty” decreases, thus, having a less effect on the heat transfer coefficient.

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Good, with the analysis in the discussion sections. But you also need to analysis why there're differences between theoretical values and experimental values.

9. Appendix III: Sample Calculations

Sample Calculation for Unbaffled Inner Tank Surface Heat Transfer Coefficient:

$$\frac{h_i D_t}{K} = 0.36 \left(\frac{D_i n \rho}{\mu} \right)^{2/3} \left(\frac{C_p \mu}{K} \right)^{1/3} \quad (8)$$

Sample Calculation for 610 RPM @ 37.78 °C:

- h_i = Convective heat transfer coefficient at the inner surface of the tank [=] $\left(\frac{W}{m^2 \cdot K} \right)$
- D_t = Tank Diameter [=] .2286 m
- μ = Viscosity [=] .0004 $\left(\frac{kg}{m \cdot s} \right)$
- C_p = Specific heat of fluid inside vessel [=] 4185.5 $\left(\frac{J}{kg \cdot s} \right)$
- K = Thermal conductivity of fluid inside vessel [=] .665 $\left(\frac{W}{m \cdot K} \right)$
- ρ = Density of fluid inside vessel [=] 1000 $\frac{kg}{m^3}$
- D_i = Impeller diameter [=] .0889 m
- n = Speed of impeller rotation [=] 610 RPM * $\left(\frac{1 \frac{rev}{s}}{60 RPM} \right) \rightarrow 10.17 \frac{revolution}{second} \left[\frac{rev}{s} \right]$

Using equation 2 and plugging all the variables in, the following is calculated:

$$h_i = 0.36 \left[\frac{(0.0889 m)^2 \left(10.17 \frac{rev}{s} \right) \left(1000 \frac{kg}{m^3} \right)}{0.0004 \frac{kg}{m \cdot s}} \right]^{2/3} \left[\frac{\left(4185.5 \frac{J}{kg \cdot K} \right) \left(0.0004 \frac{kg}{m \cdot s} \right)}{0.665 \frac{W}{m \cdot K}} \right]^{1/3} \left[\frac{.665 \frac{W}{m \cdot K}}{.2286 m} \right]$$

$$h_i = 4961.13 \frac{W}{m^2 \cdot K}$$

Commented [DL43]: 8/10
Heat balances are needed.

Commented [DL44]: Good.

Reynolds Number Sample Calculation:

$$Re = \frac{D^2 n \rho}{\mu} \quad (9)$$

$$Re = \frac{(0.0889)^2 (10.17 \frac{rev}{s}) (1000 \frac{kg}{m^3})}{.0004 \frac{kg}{m * s}}$$

$$Re = 200939.11$$

Commented [DL45]: Use the density at 37.78 C.

Prandtl Number Sample Calculation:

$$Pr = \frac{c_p \mu}{k} \quad (10)$$

$$Pr = \frac{(4185.5 \frac{J}{kg * s}) (0.0004 \frac{kg}{m * s})}{.665 \frac{W}{m * K}}$$

$$Pr = 2.517$$

Nusselt Number Sample Calculation:

$$Nu = 0.36(Re)^{2/3}(Pr)^{1/3} \quad (11)$$

$$Nu = 0.36(200939.11)^{2/3}(2.517)^{1/3}$$

$$Nu = 1680.12$$

10. Appendix IV: Individual Team ContributionsNAME: *Mahamat Abdraman*

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present in both sessions
Pre-Lab Editing	1	Editing
Experimental Objective (Prelab)	2	
Apparatus (prelab)	1	Wrote the apparatus
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)	2	Wrote safety
Final Lab Editing	1	Editing
Executive Summary (final lab)		
Introduction (Final lab)	3	Wrote introduction
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)	.5	Added references
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	Worked on PowerPoint
Total	17.5	

NAME: *Gewargis Garmooos*

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present in the lab
Pre-Lab Editing	1	Editing
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	2	Wrote procedure
Project Safety Evaluation (prelab)		
Final Lab Editing	2	Editing
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	5	Worked on the result
Discussion (final lab)	2	Worked on discussion
References (final and/or prelab)	.5	Added references
Data Tabulation/Graphs (final)	4	Recorded the data
Error Analysis (final lab)	2	Worked on writing error analysis
Sample Calculations (final Lab)	4	Worked on calculation
Power Point Presentation		
Total	29.5	

NAME: *Marvl Odeesh*

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present both days
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)	2	Wrote material and supplies
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	Editing
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	5	Worked on the result
Discussion (final lab)	2	Worked on discussion
References (final and/or prelab)	.5	Added
Data Tabulation/Graphs (final)	5	References
Error Analysis (final lab)	2	Worked on errors
Sample Calculations (final Lab)	5	Helped with calculation
Power Point Presentation		
Total	29.5	

NAME: *Sinan Masoud*

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present both days
Pre-Lab Editing	1	Editing
Experimental Objective (Prelab)	3	Wrote the objective
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	2	Editing
Executive Summary (final lab)	3	Wrote the summary
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)	.5	Added references
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	Worked on power point
Total	17.5	

NAME: Yohannes Teweldeberhan

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Data collecting
Pre-Lab Editing	1	Editing
Experimental Objective (Prelab)		
Apparatus (prelab)	2	Labelling and description
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	2	Editing
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)	4	Wrote the theory
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)	.5	Added references
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	Worked on the power point
Total	17.5	

Signature of Team Member	Date
Marvel Odeesh	2/19/2020
Gewargis Garmooos	2/19/2020
Mahamat Abdraman	2/19/2020
Sinan Masoud	2/19/2020
2/19/2020	2/19/2020